

ESAT CAMP MOCK A

Full sitting

EC-FULL-A

MATHEMATICS 1

40 MINUTES

MATHEMATICS 2

40 MINUTES

PHYSICS

40 MINUTES

CHEMISTRY

40 MINUTES

BIOLOGY

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin**.

This paper contains **5 modules** (Mathematics 1, Mathematics 2, Physics, Chemistry, Biology). Each module has **27 multiple-choice questions** and is designed for **40 minutes** (about **200 minutes** in total if you sit every module).

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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ESAT CAMP · Independent preparation resource

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Mathematics 1

- 1 A bag contains n green counters and $(n - 2)$ red counters, where $n \geq 3$.

Two counters are drawn at random from the bag without replacement.

The probability that the two counters are of different colours is $\frac{8}{15}$.

What is the sum of all possible values of n ?

- A 4
- B 6
- C 8
- D 10
- E 12
- F 16
- G 24

- 2 A set of 200 quadratic equations is considered. Each equation has exactly two real roots, and none of the roots are zero.

It is found that:

- 30% of the equations have two positive roots.
- 45% of the equations have two negative roots.

The remaining equations each have one positive and one negative root.

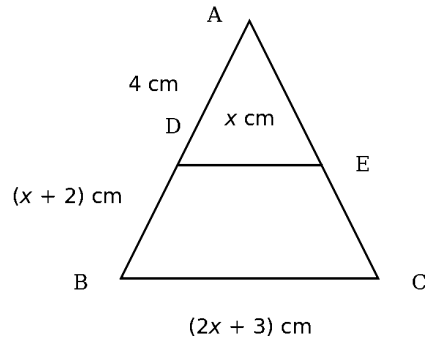
What is the total number of positive roots and negative roots, respectively, in the set?

- A (110, 140)
- B (110, 230)
- C (120, 180)
- D (170, 140)
- E (170, 230)
- F (230, 170)

- 3 In the triangle ABC shown below, points D and E lie on the sides AB and AC respectively, such that DE is parallel to BC .

$AD = 4$ cm, $DB = (x + 2)$ cm, $DE = x$ cm, and $BC = (2x + 3)$ cm.

What is the length, in cm, of BC ?



[diagram not to scale]

- A $1 + \sqrt{13}$
 B $1 + 2\sqrt{13}$
 C $3 + \sqrt{21}$
 D $5 + 2\sqrt{13}$
 E 9
 F $9 + 2\sqrt{21}$
- 4 The table shows summary statistics for an original dataset of 25 values and a subset of 5 values that is removed from it.

	number of values	mean	range
original dataset	25	40	15
removed subset	5	32	8

What can be deduced about the mean and range of the remaining 20 values?

- A mean = 38, range ≤ 15
 B mean = 38, range ≥ 8
 C mean = 38, range ≥ 15
 D mean = 42, range ≤ 15
 E mean = 42, range ≥ 8
 F mean = 42, range ≥ 15

- 5 A combined set of data consists of two groups, P and Q. The number of values in group Q is three times the number of values in group P. The mean of group P is 20 and the mean of the combined set is 32.
What is the mean of group Q?

A 28
B 36
C 44
D 68

- 6 The diagram shows a trapezium $ABCD$ in which AD is parallel to BC , and $\angle DAB = \angle ABC = 90^\circ$.

The lengths of the sides, in centimetres, are:

- $AD = x + 1$
- $AB = 3x - 1$
- $BC = 2x + 4$
- $CD = 3x + 1$

[diagram not to scale]

What is the area, in cm^2 , of the trapezium?

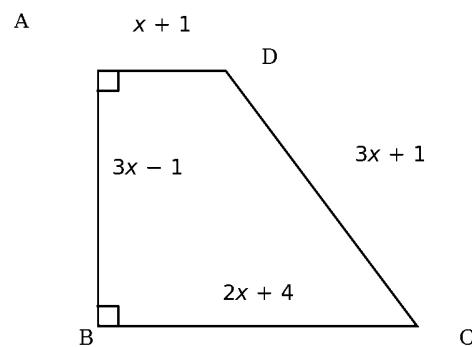


Diagram not to scale

A 24
B 32
C 48
D 56
E 64
F 72
G 112

- 7** The theoretical power available to a wind turbine is directly proportional to the cube of the wind speed.
- At a wind speed of v , the turbine operates at an efficiency of 20% and produces an actual power output of 50 kW.
- At a wind speed of $2v$, the turbine operates at an efficiency of 15%.
- What is the actual power output of the turbine at the wind speed of $2v$?
- A** 60 kW
B 75 kW
C 150 kW
D 300 kW
E 400 kW
F 1700 kW
- 8** A set of numbers, S_1 , consists of n_1 values, each equal to 10.
 A second set of numbers, S_2 , consists of n_2 values, each equal to 30.
- When the numbers from S_1 and S_2 are combined, the mean of the resulting set is 14.
- A new set is formed by combining the numbers from S_1 with a set of $2n_2$ values, each equal to 30.
- What is the mean of this new set?
- A** 18
B $\frac{50}{3}$
C 22
D $\frac{130}{7}$
E $\frac{250}{9}$
F $\frac{70}{3}$
- 9** The curve with equation $y = ax^2 + bx + c$ has a maximum value of 9 at $x = 2$. The curve passes through the point $(0, 1)$.
- What is the value of a ?
- A** -4
B -2
C -1
D 1
E 2
F 4

10 A fair six-sided die with faces numbered 1 to 6 is rolled once.

- If the number rolled is prime, a biased coin with $P(\text{Heads}) = \frac{2}{3}$ is tossed 2 times.
- If the number rolled is not prime, a fair coin is tossed 3 times.

What is the probability that exactly two heads are obtained in total?

- A** $\frac{41}{144}$
- B** $\frac{43}{108}$
- C** $\frac{91}{216}$
- D** $\frac{59}{144}$
- E** $\frac{7}{16}$
- F** $\frac{59}{72}$
- G** $\frac{3}{4}$

11 A polygon has n sides.

The mean of 5 of its interior angles is 120° .

The mean of its remaining interior angles is 160° .

What is the value of n ?

- A** 6
- B** 8
- C** 9
- D** 10
- E** 18
- F** 48

- 12 The equation

$$\left(\frac{x \times 10^3 + 4x \times 10^2}{7 \times 10^{-1}} \right)^2 - \left(\frac{x \times 10^4 + 2x \times 10^3}{3 \times 10^{-2}} \right) = 2.4 \times 10^7$$

has two real solutions for x .

What is the positive difference between these two solutions?

- A** 0.1
B 2.4
C 2.5
D 4.9
E 9.8
F 49
- 13 A composite material is formed from two substances, X and Y. Substance X has a density of 3.0 g cm^{-3} and constitutes $\frac{2}{3}$ of the material by volume. Substance Y has a density of 9.0 g cm^{-3} and constitutes the remaining $\frac{1}{3}$ by volume. What is the density of the composite material?
- A** 4.0 g cm^{-3}
B 5.0 g cm^{-3}
C 6.0 g cm^{-3}
D 7.0 g cm^{-3}
E 8.0 g cm^{-3}
- 14 Water flows through a cylindrical pipe of internal diameter 5 cm into a cylindrical tank of internal diameter 2 m.
 The depth of the water in the tank increases at a constant rate of 3 cm per minute.
 What is the speed of the water in the pipe, in m s^{-1} ?
- A** 0.02
B 0.2
C 0.8
D 3.2
E 48
F 80

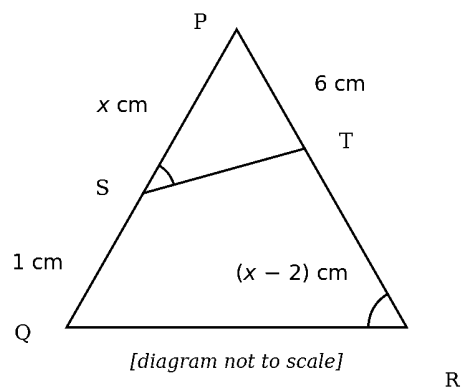
- 15 In the diagram, S is a point on side PQ and T is a point on side PR of triangle PQR .

The angle $\angle PST$ is equal to $\angle PRQ$.

The lengths of the segments are:

- $PS = x$ cm
- $SQ = 1$ cm
- $PT = 6$ cm
- $TR = (x - 2)$ cm

What is the length of side PR ?



- A 6 cm
- B 8 cm
- C 9 cm
- D 12 cm
- E $(5 + \sqrt{7})$ cm

- 16 The real numbers x and y satisfy the equation

$$2x^2 - 3xy + 2y^2 = 7$$

What is the maximum possible value of $x - y$?

- A $\frac{\sqrt{14}}{2}$
- B 2
- C $\sqrt{7}$
- D 4
- E $2\sqrt{7}$
- F $4\sqrt{2}$

- 17** A machine extrudes a continuous solid cylinder of plastic of diameter 4 cm. The volume of plastic extruded is 240π cm³ per minute.

What is the speed at which the cylinder emerges from the machine, in mm per second?

- A** 1
- B** 2.5
- C** 10
- D** 40
- E** 60
- F** 600

- 18** Seven distinct numbers are written in strictly increasing order:

$$x_1 < x_2 < x_3 < x_4 < x_5 < x_6 < x_7$$

Their mean is M , their median is m , and their range is R .

The smallest and largest of these seven numbers are then removed, leaving five numbers.

Which of the following statements must be true?

- 1 The median of the remaining five numbers is equal to m .
- 2 The range of the remaining five numbers is strictly less than R .
- 3 If $M = m$, the mean of the remaining five numbers is equal to M .

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 19** The diagram shows a right-angled trapezium with parallel horizontal sides of lengths $2x$ and $3x$, and vertical height h . The area of this trapezium is A_1 .

A new trapezium is formed by modifying these dimensions:

- the side of length $3x$ is increased by 40%
- the side of length $2x$ is decreased by 35%
- the vertical height h is increased by 20%

The area of the new trapezium is A_2 .

The relationship between A_1 and A_2 can be expressed as $A_2 = kA_1$.

What is the value of k ?

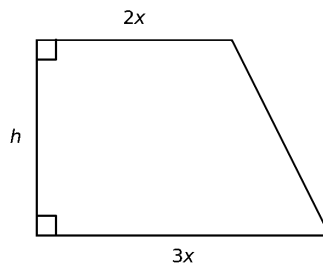


Diagram not to scale

- A** $\frac{27}{25}$
- B** $\frac{57}{50}$
- C** $\frac{5}{4}$
- D** $\frac{33}{25}$
- E** $\frac{7}{5}$

- 20** The tensile strengths of a batch of 50 manufactured fibres are tested. The batch consists of 30 carbon fibres and 20 glass fibres.

Statistics for the carbon fibres and for the entire batch of 50 fibres are shown in the table:

Fibre group	Number of fibres	Mean tensile strength / GPa	Range of tensile strength / GPa
Carbon fibres	30	3.6	1.1
All 50 fibres combined	50	3.2	1.6

What can be deduced about the mean and range of the tensile strength of the 20 glass fibres?

- A** mean = 2.6 GPa, range $\leq$ 1.6 GPa
B mean = 2.6 GPa, $1.1 \text{ GPa} \leq \text{range} \leq 1.6 \text{ GPa}$
C mean = 2.6 GPa, range $\geq$ 1.6 GPa
D mean = 2.8 GPa, range $\leq$ 1.6 GPa
E mean = 2.8 GPa, $1.1 \text{ GPa} \leq \text{range} \leq 1.6 \text{ GPa}$
F mean = 2.8 GPa, range $\geq$ 1.6 GPa

- 21** The region R in the xy -plane is defined by the inequalities:

$$x \geq 0, \quad y \geq 0, \quad x + y \leq 6, \quad 3x + y \leq 12$$

What is the area of R ?

- A** 3
B 9
C 12
D 15
E 18

- 22** The following six numbers are given in ascending order:

$$10, 16, h, 2h, 38, 44$$

The mean of these six numbers is equal to their median.

What is the value of h ?

- A** 7.2
B 12.0
C 13.5
D 18.0
E 24.0
F 36.0

- 23** It takes time T for a single pipe to fill a particular cylindrical tank.

A new cylindrical tank is used, which has three times the radius and one-third the height of the original tank.

This new tank is filled using 6 pipes simultaneously. Each of these pipes has a radius that is twice the radius of the original pipe, and the water flows through each at half the speed of the water in the original pipe.

In terms of T , what is the time taken to fill the new tank?

- A** $\frac{1}{12}T$
- B** $\frac{1}{8}T$
- C** $\frac{1}{4}T$
- D** $\frac{1}{2}T$
- E** $\frac{3}{4}T$

- 24** Find the set of values of the real constant k for which the line with equation $y = 2x$ does not intersect the circle with equation $x^2 + y^2 - 4x + 6y + k = 0$.

- A** $16/5 < k < 13$
- B** $k > 16/5$
- C** $k < 13$
- D** $k < 16/5$
- E** $k > 13$
- F** $16/5 \leq k < 13$

- 25** A square has side length $2a$.

A circle, C , is inscribed in the square.

A quarter-circle, Q , has its centre at one vertex of the square and has a radius equal to the side length of the square.

What is the area of Q minus the area of C ?

- A** 0
- B** $2\pi a^2$
- C** $3\pi a^2$
- D** $-3\pi a^2$
- E** $a^2(4 - \pi)$
- F** πa^2

- 26 Given that $p = 2^x$ and $q = 3^x$, which one of the following is an expression for

$$\frac{6^{x+1} \times 12^{x-1}}{18^{2x-1}}$$

in terms of p and q ?

- A $\frac{p}{q^2}$
 B $\frac{4p}{q^2}$
 C $\frac{9p}{q^2}$
 D $\frac{9p}{q}$
 E $\frac{p}{9q^2}$
 F $9pq^2$
- 27 The diagram shows four identical shaded rectangles arranged inside a large square to leave a central unshaded square.

The side length of the outer square is $\sqrt{15} + \sqrt{5}$.

The side length of the inner unshaded square is $\sqrt{15} - \sqrt{5}$.

What is the total area of the shaded region?

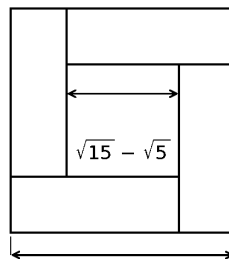


Diagram not to scale

- A 0
 B $5\sqrt{3}$
 C $10\sqrt{3}$
 D $20\sqrt{3}$
 E 40
 F $40 + 20\sqrt{3}$
 G $40\sqrt{3}$

Mathematics 2

- 1 The sum of the first 3 terms of a geometric progression is 8. The sum of the first 6 terms of the same progression is 72.
What is the common ratio of the progression?

A 2
B 3
C 4
D 8
E 9

- 2 The diagram shows a triangle with side lengths $x + 3$, $2x - 1$, and $11 - x$.
Find the complete set of real values of x for which a non-degenerate triangle with these side lengths exists.

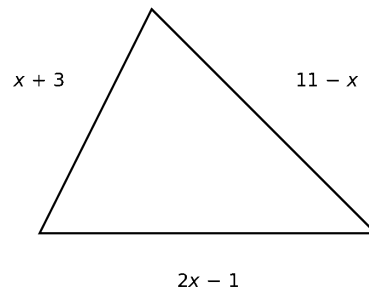


Diagram not to scale

A $x > \frac{9}{4}$
B $x < \frac{15}{2}$
C $\frac{1}{2} < x < 11$
D $\frac{9}{4} < x < \frac{15}{2}$
E $\frac{9}{4} < x < 11$
F $\frac{1}{2} < x < \frac{15}{2}$
G $\frac{1}{2} < x < \frac{9}{4}$
H $\frac{15}{2} < x < 11$

- 3 The diagram shows a regular hexagon $ABCDEF$ with sides of length a .

A circular arc with centre A is drawn from vertex C to vertex E .

What is the area of the shaded region bounded by the line segments CD and DE and the arc CE ?

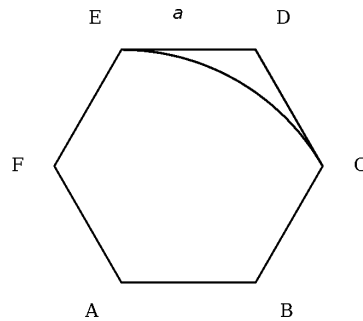


Diagram not to scale

- A $\left(\frac{\sqrt{3} - \pi}{2} \right) a^2$
- B $\left(\frac{2\sqrt{3} - \pi}{4} \right) a^2$
- C $\left(\frac{2\sqrt{3} - \pi}{2} \right) a^2$
- D $\left(\frac{3\sqrt{3} - \pi}{2} \right) a^2$
- E $\left(\frac{4\sqrt{3} - \pi}{4} \right) a^2$
- F $\left(\frac{3\sqrt{3} - 2\pi}{4} \right) a^2$

- 4 What is the value of the definite integral

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (3 + \cos x + x \cos x) \, dx$$

- A 2π
- B 3π
- C $3\pi + 1$
- D $3\pi + 2$
- E 4π

- 5 A parabola has the equation $y = x(k - x)$ for some constant $k > 0$.
A triangle is formed by the two points where the parabola meets the x -axis and by the vertex of the parabola.

What is the total area of the regions that lie between the parabolic arc and the sides of the triangle?

- A $\frac{k^3}{24}$
- B $\frac{k^3}{12}$
- C $\frac{k^3}{8}$
- D $\frac{k^3}{6}$
- E $\frac{k^3}{4}$

- 6 Find the area of the finite region defined by the inequalities:

$$y \geq |2x + 4|$$

$$y \leq 8$$

- A 8
- B 16
- C 24
- D 32
- E 48

- 7 Find the sum of the squares of the real roots of the equation

$$x\sqrt{1} = \frac{1}{3}$$

- A $\frac{1}{9}$
- B $\frac{1}{3}$
- C $\frac{7}{9}$
- D 1
- E 2

- 8 A company's annual profit is modelled to decrease by a fixed percentage, $p\%$, at the end of each year.

The total profit predicted for the first two years is 36% of the total profit predicted if the company were to operate forever under these conditions.

What is the value of p ?

- A 20
- B 36
- C 40
- D 60
- E 64
- F 80

- 9 A bag contains 7 gold tokens and 5 silver tokens. The tokens are identical apart from their colour.

A player draws tokens one at a time at random from the bag without replacement.

The player begins with a score of 0.

- Each gold token drawn increases the score by 1 point.
- When a silver token is drawn, the score decreases by 2 points and the game ends immediately.
- If all 7 gold tokens are drawn without drawing a silver token, the game also ends.

What is the probability that the player finishes the game with a score strictly greater than 0?

- A $\frac{35}{396}$
- B $\frac{7}{99}$
- C $\frac{7}{44}$
- D $\frac{5}{22}$
- E $\frac{7}{22}$
- F $\frac{5}{11}$

- 10 A particle undergoes three successive displacements, $\mathbf{d}_1$, $\mathbf{d}_2$, and $\mathbf{d}_3$, and returns to its starting position. $\mathbf{d}_1$ has magnitude $\sqrt{3}$ units and a bearing of 090° . $\mathbf{d}_2$ has magnitude 1 unit and a bearing of 180° .

What is the bearing of $\mathbf{d}_3$?

- A 060°
- B 120°
- C 240°
- D 300°
- E 330°

- 11 What is the coefficient of x^{24} in the expansion of the expression below?

$$((1 - 2x^2)(1 + 2x^2 + 4x^4 + 8x^6))^5$$

- A -245760
- B -40960
- C -80
- D 0
- E 20480
- F 40960

- 12 The definite integral of a function $f(x)$ over the interval $0 \leq x \leq 6$ is given by $\int_0^6 f(x) \, dx = -12$.

The area of the region bounded by the curve $y = f(x)$, the x -axis, and the lines $x = 0$ and $x = 6$ that lies above the x -axis is 5.

What is the area of the region bounded by the curve, the x -axis, and the lines $x = 0$ and $x = 6$ that lies below the x -axis?

- A 5
- B 7
- C 12
- D 17
- E 22

- 13 What is the value of the following sum?

$$\sum_{n=1}^{63} \log_4 \left(\frac{n^2 + n}{n^2 + 2n + 1} \right)$$

- A -6
- B -3
- C 0
- D 3
- E 6

- 14 A straight line with a negative gradient passes through the point $(4, 3)$.

The line forms a triangle with the positive x -axis and positive y -axis.

The line $y = 2x$ divides this triangle into two smaller triangles.

The area of the triangle with a side on the x -axis is three times the area of the triangle with a side on the y -axis.

What is the area of the triangle formed by the straight line and the positive coordinate axes?

- A $\frac{289}{12}$
- B 27
- C $\frac{121}{4}$
- D $\frac{75}{2}$
- E $\frac{289}{6}$

- 15 How many distinct real solutions does the following equation have?

$$\ln(x - 5) + \ln(2 - x) = \ln(2)$$

- A 0
- B 1
- C 2
- D 3
- E 4
- F 7

- 16 A sector of a circle has a fixed perimeter P .

Which expression gives the radius r that maximises the area of the sector?

- A $\frac{P}{2+\pi}$
- B $\frac{P}{4}$
- C $\frac{P}{\pi}$
- D $\frac{P}{3}$
- E $\frac{P}{2}$

- 17 A sequence is defined by $u_1 = 2$ and the recurrence relation

$$u_{n+1} = \ln(e^{u_n-1} - k)$$

for $n \geq 1$, where k is a real constant.

The sequence terminates if $e^{u_n-1} - k \leq 0$.

Find the set of values of k for which the sequence has exactly two strictly positive terms.

- A** $0 \leq k < e - 1$
B $0 \leq k < \frac{e}{e+1}$
C $0 < k \leq e - 1$
D $0 \leq k \leq e - 1$
E $\frac{e^2 - 1}{2} \leq k < e^2 - 1$
- 18 Two rectangles, A and B, have dimensions that depend on a variable x , which changes with time. The side lengths of rectangle A are $(\log_2(8) + 3x)$ and $(12 - x)$. The side lengths of rectangle B are $(\log_3(27) + x)$ and $(6 - x)$. At a certain instant, the perimeter of rectangle A is 42. At this instant, x is decreasing. What is the value of the ratio $\frac{\text{Area of A}}{\text{Area of B}}$ at this instant, and is this ratio increasing or decreasing?
- A** The ratio is 3 and it is decreasing.
B The ratio is 3 and it is increasing.
C The ratio is 6 and it is decreasing.
D The ratio is 6 and it is increasing.
E The ratio is $\frac{1}{6}$ and it is decreasing.
F The ratio is $\frac{1}{6}$ and it is increasing.
- 19 Find the x -coordinate of the stationary point of the curve with equation:

$$y = \frac{x^2 - 4x}{x^2 - 16}(x + 4)^2$$

- A** -4
B -2
C 0
D 2
E 4

- 20** A particle is dropped from a height of 10 m. After each bounce, it reaches a maximum height that is a constant fraction, r , of the maximum height of the previous bounce, where $0 < r < 1$.

The particle comes to rest after travelling a total vertical distance of 50 m.

What is the value of r ?

- A** $r = \frac{1}{2}$
- B** $r = \frac{3}{5}$
- C** $r = \frac{2}{3}$
- D** $r = \frac{5}{7}$
- E** $r = \frac{4}{5}$

- 21** Which of the following statements is/are correct?

1. $2^{\log_3 5} = 5^{\log_3 2}$

2. $\frac{\sqrt{75} - \sqrt{12}}{\sqrt{27}} = 1$

3. $\frac{\log_{10} 8}{\log_{10} 2} = \log_{10} 4$

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 22** Find the sum of all solutions to the equation

$$\ln(-\sin x) + \ln(-\cos x) = -\ln 4$$

in the interval $0 \leq x \leq 2\pi$.

- A** 0
- B** $\frac{\pi}{2}$
- C** π
- D** $\frac{5\pi}{2}$
- E** 3π

- 23** The index, $I(f)$, of a polynomial $f(x)$ is defined by the formula:

$$I(f) = d + s + |f(-1)|$$

where d is the degree of $f(x)$, and s is the number of distinct real stationary points of $f(x)$.

Consider the two polynomials:

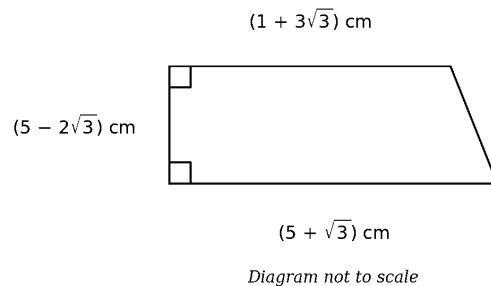
$$f_1(x) = x^3 - 3x + 1$$

$$f_2(x) = x^3 - 5$$

What is the value of $I(f_1) + I(f_2)$?

- A** 6
 - B** 8
 - C** 16
 - D** 17
 - E** 18
 - F** 20
- 24** A rock sample contains a radioactive isotope X and its stable decay product Y. The ratio of the number of nuclei of X to the number of nuclei of Y is 1:3.
The half-life of X is T.
Assuming that there was no Y in the sample initially and that all nuclei of Y were produced by the decay of X, which of the following is the age of the sample?
- A** 1.5 T
 - B** 2 T
 - C** 3 T
 - D** 4 T
 - E** T

- 25** The diagram shows a right-angled trapezium with parallel horizontal sides of length $(1 + 3\sqrt{3})$ cm and $(5 + \sqrt{3})$ cm, and a perpendicular height of $(5 - 2\sqrt{3})$ cm.
- What is the area of the trapezium?



- A** $(3 + 4\sqrt{3}) \text{ cm}^2$
- B** $(6 + 8\sqrt{3}) \text{ cm}^2$
- C** $(27 + 4\sqrt{3}) \text{ cm}^2$
- D** $(21 - 4\sqrt{3}) \text{ cm}^2$
- E** $(3 + 16\sqrt{3}) \text{ cm}^2$
- F** $(9 + 2\sqrt{3}) \text{ cm}^2$
- 26** A sequence is defined by the general term

$$u_n = \frac{9^{n+2}}{3^{2n+1}}$$

for integers $n \geq 1$.

What is the value of the sum of the first 10 terms of the sequence?

- A** $\frac{10}{27}$
- B** 27
- C** 30
- D** 270
- E** 810
- 27** Find the area of the finite region bounded by the curve with equation $4y + |3x| = 24$ and the x -axis.
- A** 24
- B** 48
- C** 96
- D** 144
- E** 192
- F** 288

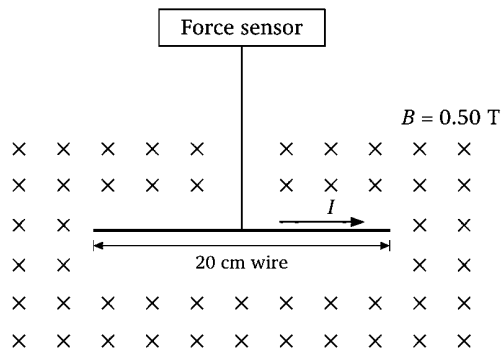
Physics

- 1 A point on a guitar string oscillates in simple harmonic motion with a frequency of 50 Hz and an amplitude of 4.0 mm.

What is the total distance travelled by this point during a time interval of 3.0 s?

- A 0.0 m
- B 0.6 m
- C 1.2 m
- D 2.4 m
- E 3.8 m

- 2 A straight, horizontal wire of length 20 cm is suspended from a sensitive force sensor. The wire is in a uniform magnetic field of strength 0.50 T, directed perpendicular to the wire. A total charge of 12 C passes through the wire over a time of 2.0 s. What is the magnitude of the change in the reading on the force sensor?



- A 0.20 N
- B 0.60 N
- C 1.2 N
- D 2.4 N
- E 3.0 N
- F 60 N

- 3 Two objects, X and Y, are in motion. Object X has 3 times the momentum of object Y and 4 times the kinetic energy of object Y.

What is the ratio of their masses, $\frac{m_X}{m_Y}$?

- A $\frac{3}{16}$
- B $\frac{3}{4}$
- C $\frac{4}{9}$
- D $\frac{9}{4}$
- E 12
- F 36

- 4 The intensity I of a physical field varies with distance r from a point source according to a power law relationship $I = kr^n$, where k and n are constants. Experimental measurements yield the following data:

$$\text{At } r = 2.0 \text{ m, } I = 72 \text{ W m}^{-2}$$

$$\text{At } r = 4.0 \text{ m, } I = 18 \text{ W m}^{-2}$$

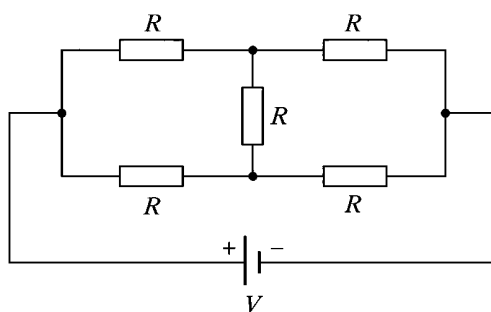
What is the intensity I at a distance of $r = 3.0 \text{ m}$?

- A 24 W m^{-2}
- B 32 W m^{-2}
- C 36 W m^{-2}
- D 45 W m^{-2}
- E 48 W m^{-2}

- 5 A fixed mass of gas is sealed in a rigid container and heated. Which statement best explains the increase in pressure?

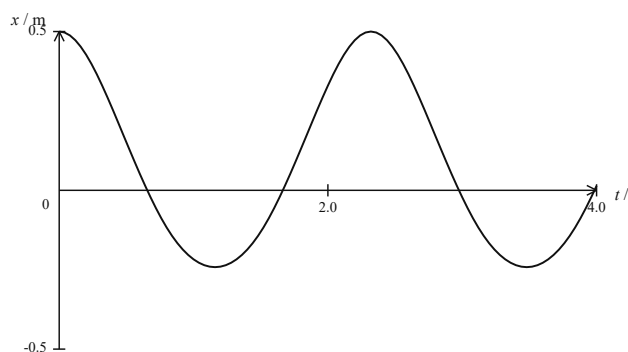
- A The particles have greater average speed and transfer more momentum to the walls per unit time.
- B The particles become larger, so less empty space remains.
- C The number of particles increases while their average speed remains constant.
- D The particles move more slowly and therefore remain in contact with the walls for longer.
- E The average spacing between particles must increase because the gas is hotter.

- 6 A transmission line receives 20 kW at 2.0 kV. The total resistance of its cables is 5.0Ω . The voltage at the sending end is increased to 10 kV, while the power entering the line remains 20 kW. The cable resistance is unchanged. By how much does the power lost in the cables decrease?
- A 20 W
B 100 W
C 400 W
D 480 W
E 500 W
- 7 A circuit is constructed with five identical resistors, each of resistance R , connected to a DC supply of voltage V . The circuit consists of two parallel branches, each containing two resistors in series. A fifth identical resistor connects the midpoint of the first branch to the midpoint of the second branch. The total power dissipated by the circuit is P . If the connecting resistor between the two branches is removed, which expression gives the new total power dissipated?



- A $\frac{1}{2}P$
B $\frac{4}{5}P$
C P
D $\frac{5}{4}P$
E $2P$

- 8 The graph shows the displacement of a particle undergoing simple harmonic motion as a function of time.



What is the average speed of the particle during the first 4.0 s of its motion?

- A 0 m s^{-1}
 - B 0.25 m s^{-1}
 - C 0.50 m s^{-1}
 - D 1.0 m s^{-1}
 - E $\frac{\pi}{2} \text{ m s}^{-1}$
 - F 2.0 m s^{-1}
- 9 Three devices use different regions of the electromagnetic spectrum:
1. a thermal camera detecting radiation emitted by a warm hand;
 2. a lamp producing ultraviolet radiation;
 3. a scanner producing X-rays.

Which row lists the devices in order of increasing radiation frequency and correctly compares the speeds of their radiation in a vacuum?

- A 1, 2, 3; all three speeds are equal
- B 1, 3, 2; all three speeds are equal
- C 3, 2, 1; all three speeds are equal
- D 1, 2, 3; speed increases in this order
- E 2, 1, 3; speed decreases in this order

- 10** Each diagram is intended to show a ray travelling from air through a rectangular glass block and back into air. Light travels more slowly in glass than in air. The two faces crossed are parallel. Which diagram shows a possible path? Ignore partial reflection.

- A** P
- B** Q
- C** R
- D** S
- E** None of them

- 11** Straight water waves cross a boundary from deep water into shallow water. In the deep water, the distance from the first crest to the fifth crest is 24 cm. The wave speed there is 30 cm s^{-1} .

In the shallow water, the distance from the first crest to the fifth crest is 16 cm.

What are the wave frequency and the speed in the shallow water?

- A** 1.25 Hz, 20 cm s^{-1}
- B** 5.0 Hz, 20 cm s^{-1}
- C** 5.0 Hz, 45 cm s^{-1}
- D** 7.5 Hz, 30 cm s^{-1}
- E** 1.25 Hz, 5.0 cm s^{-1}

- 12** An electric kettle has a power input of 2.5 kW. When used to boil water, it produces steam at a steady rate of 1.0 g s^{-1} .
- What is the rate of energy loss from the kettle to the surroundings?
- (Specific latent heat of vaporisation of water = $2.0 \times 10^6 \text{ J kg}^{-1}$; specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$)
- A** 0 W
B 500 W
C 1500 W
D 2000 W
E 2500 W
F 4500 W
- 13** A nucleus of Thorium-232 (${}_{90}^{232}\text{Th}$) decays through a series of emissions to become a stable nucleus of Lead-208 (${}_{82}^{208}\text{Pb}$).
- The decay series consists entirely of alpha (α) particles and beta-minus (β^{-}) particles.
- What are the total numbers of α and β^{-} particles emitted during this process?
- A** 4 α , 6 β^{-}
B 6 α , 4 β^{-}
C 6 α , 8 β^{-}
D 6 α , 12 β^{-}
- 14** A small heating element at the bottom of an open tank of liquid heats a small parcel of the liquid immediately above it.
- Just before this heated parcel begins to rise, how do its mass, weight, and the upthrust acting on it compare to their values before it was heated?
- A** Mass: Unchanged, Weight: Unchanged, Upthrust: Increases
B Mass: Unchanged, Weight: Unchanged, Upthrust: Decreases
C Mass: Unchanged, Weight: Decreases, Upthrust: Unchanged
D Mass: Decreases, Weight: Decreases, Upthrust: Unchanged
E Mass: Decreases, Weight: Decreases, Upthrust: Increases
F Mass: Unchanged, Weight: Decreases, Upthrust: Increases

- 15 Two identical copper spheres, **X** and **Y**, are heated to 500 K.

Sphere **X** has a matte black surface.

Sphere **Y** has a polished silver surface.

The initial rate of energy loss, R , is determined for each sphere in a **vacuum** and in an atmosphere of **argon** gas at 298 K.

In the vacuum, it is observed that $R_X > R_Y$.

Assume that the rate of heat transfer due to convection depends only on the dimensions of the sphere and the temperature difference.

Which statement correctly describes the rates in the argon atmosphere compared to the vacuum?

- A The rates R_X and R_Y increase by the same amount, and the ratio $\frac{R_X}{R_Y}$ decreases.
- B The rates R_X and R_Y increase by the same amount, and the ratio $\frac{R_X}{R_Y}$ stays constant.
- C The rate R_X increases more than R_Y because black surfaces are better thermal conductors, and the ratio $\frac{R_X}{R_Y}$ increases.
- D The rate R_Y increases more than R_X because the polished surface reflects gas molecules, and the ratio $\frac{R_X}{R_Y}$ decreases.

- 16 A continuous wave travels from medium X to medium Y. The wave speed in medium Y is $\frac{2}{3}$ of the wave speed in medium X.

What is the fractional change in the wavelength of the wave?

- A $-\frac{1}{3}$
- B 0
- C $\frac{1}{3}$
- D $\frac{1}{2}$
- E $\frac{2}{3}$

- 17 The radioactive nuclide ${}_{92}^{235}\text{U}$ decays into the stable nuclide ${}_{82}^{207}\text{Pb}$ via a series of α (alpha) and β^- (beta-minus) emissions.

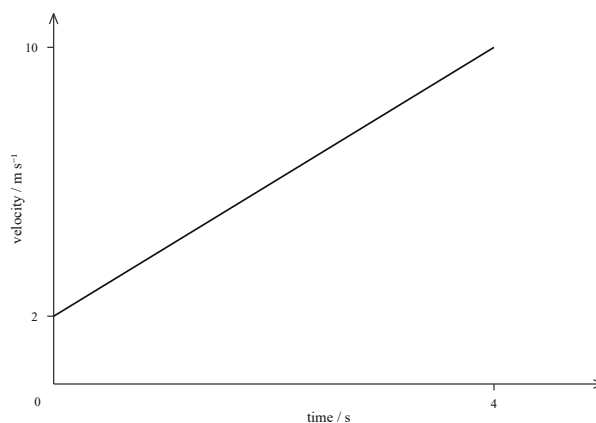
Which option gives the total number of α and β^- particles emitted during this decay chain?

- A 5 α , 0 β^-
- B 7 α , 4 β^-
- C 7 α , 10 β^-
- D 8 α , 6 β^-

- 18** An electric heater of constant power is used to heat a metal block. The block is initially at room temperature, 20°C .
 After 10 minutes, the temperature of the block is 50°C .
 The rate of thermal energy loss to the surroundings is proportional to the temperature difference between the block and the surroundings.
 Which of the following describes the temperature θ of the block after a total of 20 minutes?

- A** $\theta = 50^{\circ}\text{C}$
- B** $50^{\circ}\text{C} < \theta < 80^{\circ}\text{C}$
- C** $\theta = 80^{\circ}\text{C}$
- D** $80^{\circ}\text{C} < \theta < 100^{\circ}\text{C}$
- E** $\theta = 100^{\circ}\text{C}$

- 19** An object of mass 2.0 kg moves in a straight line. The velocity-time graph for its motion is shown.



What is the magnitude of the constant net force acting on the object?

- A** 1.0 N
- B** 2.0 N
- C** 4.0 N
- D** 5.0 N
- E** 12 N
- F** 20 N

- 20** A 12 V d.c. supply is connected to the circuit shown. The ideal diode initially conducts. A conducting ideal diode has zero resistance; in the reverse direction it blocks all current. The supply connections are reversed. What is the new total power supplied to the circuit?

- A** 6.0 W
- B** 9.0 W
- C** 12 W
- D** 18 W
- E** 36 W

- 21** The left-hand end of a solenoid is viewed directly. The current around this end flows anticlockwise, as shown. A small compass is placed on the solenoid's axis just beyond its right-hand end. Other magnetic fields are negligible. Which way does the compass's north-seeking end point initially, and after the current is reversed?

- A** right initially; right afterwards
- B** right initially; left afterwards
- C** left initially; left afterwards
- D** left initially; right afterwards
- E** upwards initially; downwards afterwards

- 22** Twelve identical resistors are connected to form the edges of a cube. A current enters the network at one vertex and leaves at the diagonally opposite vertex.

Let P_{in} be the power dissipated in a single resistor connected directly to the input vertex.

Let P_{mid} be the power dissipated in a single resistor that is connected to neither the input nor the output vertex.

What is the value of the ratio $\frac{P_{\text{in}}}{P_{\text{mid}}}$?

- A** 1
 - B** 2
 - C** 4
 - D** 9
 - E** 16
- 23** A solid composite object has total volume $4.0 \times 10^{-3} \text{ m}^3$ and is made from materials X and Y. Their densities are 2000 kg m^{-3} and 800 kg m^{-3} respectively.
- It is fully submerged and neutrally buoyant in a fluid of density 1100 kg m^{-3} . For this question, neutral buoyancy means that the object's total mass equals the mass of the displaced fluid:
- $$m_{\text{total}} = \rho_{\text{fluid}} V_{\text{total}}.$$
- What is the mass of material X?
- A** 1.1 kg
 - B** 2.0 kg
 - C** 2.4 kg
 - D** 4.0 kg
 - E** 4.4 kg
 - F** 6.0 kg
- 24** An ideal transformer has one primary coil and two separate secondary coils. The primary has 600 turns and is supplied at 120 V a.c.
- Secondary X has 60 turns and supplies a 6.0Ω resistor. Secondary Y has 120 turns and supplies a 24Ω resistor. Both loads are connected simultaneously.
- What is the primary current?
- A** 0.20 A
 - B** 0.30 A
 - C** 0.40 A
 - D** 0.60 A
 - E** 3.0 A

- 25** A fixed mass of an ideal gas initially has pressure P and volume V . The gas is compressed at constant temperature to a final volume of $V/3$. Which expression gives the final pressure of the gas?
- A** $\frac{P}{9}$
 - B** $\frac{P}{3}$
 - C** P
 - D** $3P$
 - E** $9P$
- 26** A short pulse of light travels normally through a transparent block of thickness 12 cm. Its speed in the block is $2.0 \times 10^8 \text{ m s}^{-1}$. The block is replaced by a second block of the same thickness in which the pulse travels at $1.5 \times 10^8 \text{ m s}^{-1}$. The rest of the pulse's path is unchanged. By how much does its arrival at a detector become later?
- A** 0.10 ns
 - B** 0.20 ns
 - C** 0.40 ns
 - D** 0.60 ns
 - E** 0.80 ns
- 27** A DC power supply of voltage V and negligible internal resistance is connected in series with two resistors of resistances R and $2R$. Which expression gives the current flowing through the resistor of resistance R ?
- A** $\frac{V}{3R}$
 - B** $\frac{V}{2R}$
 - C** $\frac{2V}{3R}$
 - D** $\frac{V}{R}$
 - E** $\frac{3V}{2R}$

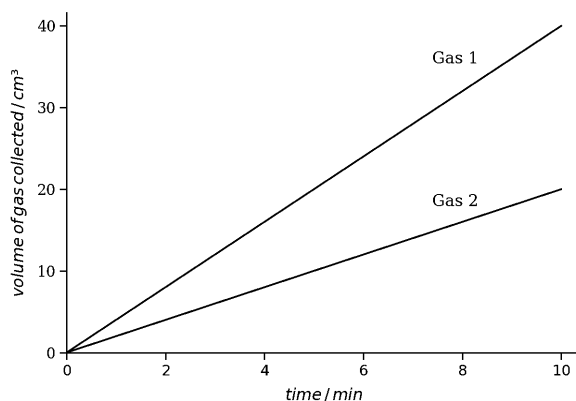
Chemistry

- 1 A constant direct electric current is passed through a dilute aqueous solution of sodium sulfate, $\text{Na}_2\text{SO}_4(\text{aq})$, using inert platinum electrodes. Gases are evolved at both electrodes.

The graph shows the volume of each gas collected over time at constant temperature and pressure.

Which of the following statements is/are correct?

1. Gas 1 is produced at the cathode (negative electrode).
2. In the external circuit, electrons flow away from the electrode producing Gas 2 towards the electrode producing Gas 1.
3. As the electrolysis proceeds, the concentration of Na_2SO_4 in the solution decreases.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 2 A metabolic pathway converts one molecule of substrate S into two molecules of product P ($1S \rightarrow 2P$). The overall process has a **net** yield of 4 molecules of ATP per molecule of S consumed.

The pathway proceeds in two sequential stages:

Stage	Event	ATP Change (per molecule of S)
1	Activation of S	Consumes 2 ATP
2	Conversion to P	Produces ATP

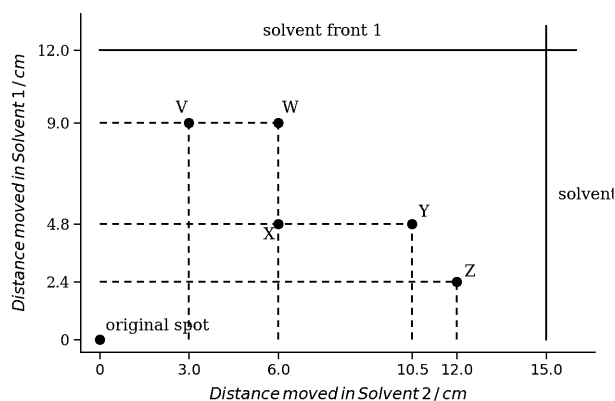
How many molecules of ATP are produced in Stage 2 for **each** molecule of P formed?

- A 1
B 2
C 3
D 4
E 6
- 3 A student carried out a two-dimensional thin-layer chromatography experiment to separate a mixture containing five organic compounds: V, W, X, Y, and Z.

A spot of the mixture was applied at the origin (0, 0). The plate was developed in Solvent 1 until the solvent front reached 12.0 cm along the vertical axis. The plate was dried, rotated by 90° , and developed in Solvent 2 until the solvent front reached 15.0 cm along the horizontal axis.

The final positions of the five compounds are shown on the chromatogram below.

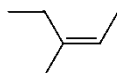
Which of the following statements is correct?



- A Developing the mixture using Solvent 1 alone would resolve all five compounds into separate spots.
B Developing the mixture using Solvent 2 alone would resolve all five compounds into separate spots.
C Compound X has the same R_f value in Solvent 1 as it has in Solvent 2.
D Compound Z has a greater R_f value in Solvent 1 than in Solvent 2.
E The R_f value of compound Y is 0.70 in Solvent 1 and 0.40 in Solvent 2.

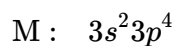
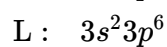
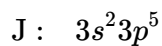
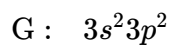
- 4 The alkene shown below reacts with hydrogen bromide, HBr, in an addition reaction:
Which of the following products is/are formed in this reaction?

- 1 $\text{CH}_3\text{CH}_2\text{CBr}(\text{CH}_3)\text{CH}_2\text{CH}_3$
2 $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CHBrCH}_3$
3 $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{Br}$



- A none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3
- 5 Element X has atomic number 4. It reacts with chlorine to form a binary compound.
Which of the following options correctly identifies the formula of this compound and describes its structure in the solid state?
- A XCl_2 , ionic lattice
B XCl_2 , simple molecular
C XCl_2 , polymeric chains
D XCl_4 , simple molecular
E XCl , ionic lattice

- 6 The ground state electronic configurations of the outer shells of four elements, denoted by the letters G, J, L, and M, are given below:



Which of the following correctly lists these elements in order of **increasing** chemical reactivity?

- A L, G, M, J
 - B G, M, J, L
 - C L, J, M, G
 - D J, M, G, L
- 7 Concentrated aqueous solutions of the five ionic compounds listed below are electrolysed using inert carbon electrodes.

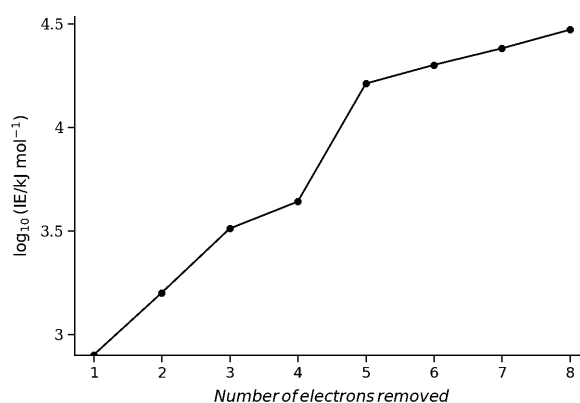
Which compound produces the elemental metal at the cathode and the elemental halogen at the anode?

- A NaCl
- B MgBr₂
- C AgF
- D CuCl₂
- E KI

- 8 The graph shows the first eight successive ionisation energies (IE) of an element Q, plotted as $\log_{10}(\text{IE}/\text{kJ mol}^{-1})$ against the number of electrons removed. Element Q belongs to Period 3 of the Periodic Table.

Which row in the table correctly identifies the Group of element Q, the formula of its stable chloride, and the total number of unpaired electrons in a ground-state neutral atom of Q?

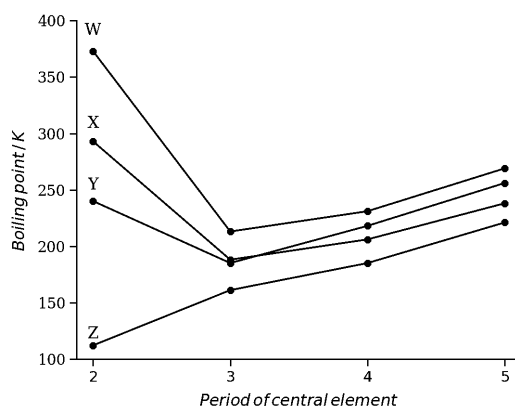
	Group	Formula of chloride	Number of unpaired electrons
A	13	QCl_3	1
B	13	QCl_3	3
C	14	QCl_4	0
D	14	QCl_4	2
E	14	QCl_4	4
F	15	QCl_5	1
G	15	QCl_5	3
H	15	QCl_3	5



- 9 The graph shows the boiling points of the hydrides of elements in Groups 14, 15, 16, and 17 for Periods 2 to 5.

Which of the following statements is/are correct?

- 1 Curve W represents the hydrides of Group 16, and the anomalously high boiling point of its Period 2 hydride is due to hydrogen bonding.
- 2 From Period 3 to Period 5 along each curve, the boiling point increases because the intramolecular covalent bonds become progressively stronger.
- 3 Curve Z represents the hydrides of Group 14, which show no anomaly in Period 2 because CH_4 molecules cannot form hydrogen bonds.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 10** An extract from a plant secretion contains three volatile liquids: P, Q, and R. A student attempts to purify each liquid using the following procedure:

1. Place the mixture in a separating funnel and allow the layers to settle.
2. Run the layers into separate flasks.
3. Perform fractional distillation on the contents of each flask.

The properties of the pure liquids are shown below:

Liquid	Density / g cm ⁻³	Boiling point / °C	Miscibility with P
P	0.79	78	-
Q	0.81	78	Miscible
R	1.00	78	Immiscible

Which of the liquids can be isolated in a pure form?

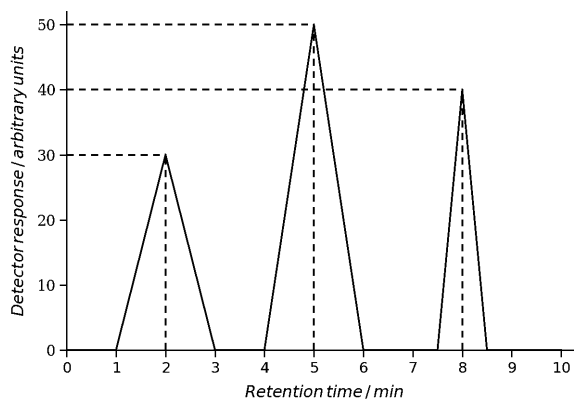
- A** P only
- B** R only
- C** P and Q only
- D** P and R only
- E** Q and R only
- F** P, Q and R
- G** None of them

- 11** A sample containing a mixture of three volatile organic compounds is analysed using gas chromatography with a non-polar stationary phase and an unreactive carrier gas as the mobile phase.

The detector response is directly proportional to the concentration of each compound, and each chromatographic peak can be approximated as a triangle (area = $\frac{1}{2} \times \text{base} \times \text{height}$).

The resulting chromatogram is shown below.

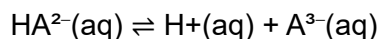
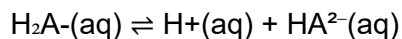
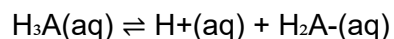
What percentage of the total peak area corresponds to the compound with the strongest attraction to the stationary phase?



- A** 20%
- B** 25%
- C** 30%
- D** 33%
- E** 40%
- F** 50%

- 12** Citric acid, whose structure is shown below, is a naturally occurring organic acid found in citrus fruits.

In aqueous solution, citric acid is a weak acid that ionises in three successive equilibrium steps:

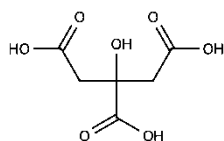


where H_3A represents citric acid.

A 0.10 mol dm^{-3} aqueous solution of hydrochloric acid, $\text{HCl}(\text{aq})$, has a pH of 1.0 at 25°C .

Which of the following statements about these solutions is/are correct?

- 1 The pH of a 0.10 mol dm^{-3} aqueous solution of citric acid at 25°C is greater than 1.0.
- 2 In aqueous solution, the dihydrogen citrate ion, H_2A^- , can act as both a Brønsted–Lowry acid and a Brønsted–Lowry base.
- 3 Exactly 30.0 cm^3 of 0.10 mol dm^{-3} barium hydroxide solution, $\text{Ba}(\text{OH})_2(\text{aq})$, is required to completely neutralise 20.0 cm^3 of 0.10 mol dm^{-3} citric acid solution.

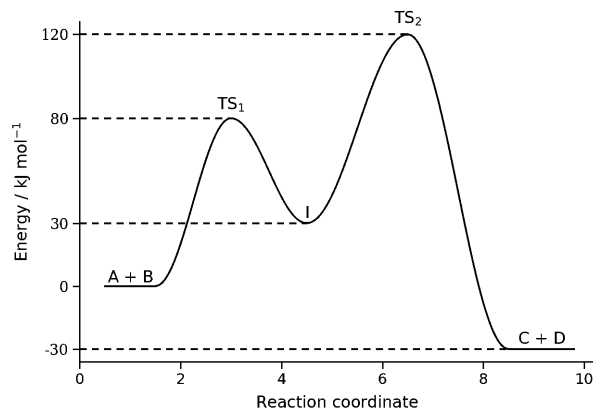


- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 13 The energy profile for a two-step reaction $A + B \rightarrow C + D$ proceeding via an intermediate I is shown in the diagram.

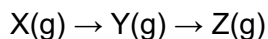
Which of the following statements is/are correct?

- 1 The overall reaction is exothermic with an enthalpy change of $\Delta H = -30 \text{ kJ mol}^{-1}$.
- 2 The activation energy for the rate-determining step of the forward reaction is $+90 \text{ kJ mol}^{-1}$.
- 3 The activation energy for the reverse reaction ($C + D \rightarrow A + B$) is $+120 \text{ kJ mol}^{-1}$.



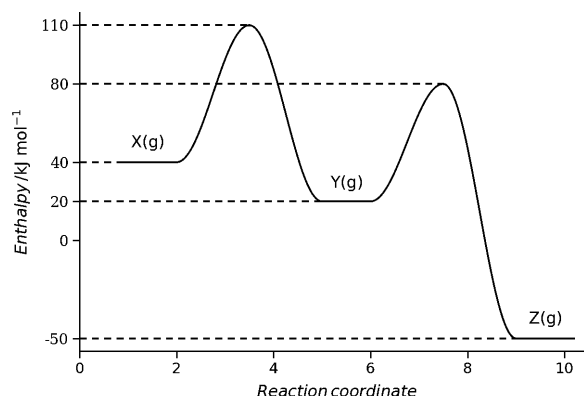
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 14 The diagram shows the enthalpy profile for a gaseous reaction that proceeds via a two-step mechanism:



Which of the following statements is/are correct?

- 1 The first step, $\text{X(g)} \rightarrow \text{Y(g)}$, is the rate-determining step of the forward reaction.
- 2 The activation energy for the reverse reaction of the first step, $\text{Y(g)} \rightarrow \text{X(g)}$, is $+70 \text{ kJ mol}^{-1}$.
- 3 The overall enthalpy change for the reaction $\text{X(g)} \rightarrow \text{Z(g)}$ is -90 kJ mol^{-1} .



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3
- 15 Element X has the atomic number 15. Which of the following correctly describes the formula and the principal type of bonding in the simplest hydride of X?
- A Formula: XH_2 ; Bonding: covalent
- B Formula: XH_3 ; Bonding: ionic
- C Formula: XH_3 ; Bonding: covalent
- D Formula: XH_4 ; Bonding: covalent
- E Formula: XH_5 ; Bonding: covalent

- 16** The ions W^{2-} , X^{-} , Y^{+} and Z^{2+} are isoelectronic (they have the same number of electrons). Which option correctly lists the **elements** W, X, Y and Z in order of increasing atomic number, and the **ions** in order of increasing ionic radius?
- A** Atomic number: $W < X < Y < Z$; Ionic radius: $Z^{2+} < Y^{+} < X^{-} < W^{2-}$
- B** Atomic number: $W < X < Y < Z$; Ionic radius: $W^{2-} < X^{-} < Y^{+} < Z^{2+}$
- C** Atomic number: $Z < Y < X < W$; Ionic radius: $Z^{2+} < Y^{+} < X^{-} < W^{2-}$
- D** Atomic number: $Z < Y < X < W$; Ionic radius: $W^{2-} < X^{-} < Y^{+} < Z^{2+}$
- 17** A 0.010 mol sample of an ionic solid is dissolved in water to form a solution. The solution contains a total of 0.050 mol of ions. Which compound could be the ionic solid?
- A** NaCl
- B** $MgCl_2$
- C** $AlCl_3$
- D** Na_3PO_4
- E** $Al_2(SO_4)_3$
- F** $C_6H_{12}O_6$
- 18** An organic compound X contains only carbon, hydrogen, and sulfur ($C_xH_yS_z$). Three separate oxidation processes are carried out using 1.0 mol of compound X in each case:
- Process 1: Complete combustion of 1.0 mol of X produces $CO_2(g)$, $H_2O(l)$, and $SO_2(g)$ only, requiring exactly 7.0 mol of $O_2(g)$.
 - Process 2: Incomplete combustion of 1.0 mol of X produces $CO_2(g)$, $H_2O(l)$, and solid sulfur, $S(s)$, only, requiring exactly 6.0 mol of $O_2(g)$.
 - Process 3: Incomplete combustion of 1.0 mol of X produces $CO(g)$, $H_2O(l)$, and solid sulfur, $S(s)$, only, requiring exactly 4.0 mol of $O_2(g)$.
- What is the molecular formula of compound X?
- A** $C_3H_6S_2$
- B** C_3H_8S
- C** C_4H_6S
- D** C_4H_8S
- E** $C_4H_8S_2$
- F** $C_4H_{10}S$
- G** C_5H_8S
- H** $C_5H_{10}S$

- 19 A researcher intends to purify organelle **Q** from a crude cellular lysate containing three major organelles: P, Q, and R. The physical properties of these organelles are shown in the table below.

Organelle	Mean Diameter (μm)	Density (g cm^{-3})
P	12.0	1.30
Q	2.5	1.15
R	0.05	1.50

The lysate is prepared in a buffer with a density of 1.05 g cm^{-3} .

Which of the following procedures represents the most effective **initial step** to purify organelle Q?

- A** Centrifuge at $1\,000g$ (pellets particles $> 10\mu\text{m}$) and retain the supernatant.
- B** Centrifuge at $10\,000g$ (pellets particles $> 2.0\mu\text{m}$) and retain the pellet.
- C** Centrifuge at $100\,000g$ (pellets particles $> 0.02\mu\text{m}$) and retain the pellet.
- D** Centrifuge at $1\,000g$ (pellets particles $> 10\mu\text{m}$) and retain the pellet.
- 20 An oxyanion has the formula $[\text{XO}_4]^{3-}$, where element X has atomic number Z and mass number A . All oxygen atoms are the isotope $^{16}_8\text{O}$.

The $[\text{XO}_4]^{3-}$ ion contains a total of E electrons and a total of N neutrons.

Which row in the table gives the correct expressions for the atomic number Z , the number of neutrons in one atom of X, and the mass number A of element X?

	atomic number of X (Z)	neutrons in one atom of X	mass number of X (A)
A	$E-29$	$N-32$	$N+E-61$
B	$E-32$	$N-32$	$N+E-64$
C	$E-35$	$N-32$	$N+E-67$
D	$E-35$	$N-32$	$N+E-64$
E	$E-35$	$N-35$	$N+E-70$
F	$E-29$	$N-29$	$N+E-58$

21 Three separate organic reactions are carried out:

Reaction 1: but-2-ene is reacted with steam in the presence of an acid catalyst.

Reaction 2: butan-2-ol is heated under reflux with acidified potassium dichromate(VI).

Reaction 3: propanoic acid is heated with ethanol in the presence of a concentrated sulfuric acid catalyst.

Which of the following statements is/are correct?

1. The organic product of Reaction 1 contains a chiral carbon atom.
2. The organic product of Reaction 2 forms a silver mirror when warmed with Tollens' reagent.
3. The organic product of Reaction 3 is a structural isomer of pentanoic acid.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

22 A student mixes 40 cm^3 of 0.75 mol dm^{-3} NaOH solution with 60 cm^3 of 1.00 mol dm^{-3} HCl solution. Both solutions are initially at 20.0°C .

The enthalpy change of neutralisation for the reaction is -56 kJ mol^{-1} .

Assume that the density of the final mixture is 1.0 g cm^{-3} and the specific heat capacity is $4.2 \text{ J g}^{-1} \text{ K}^{-1}$.

What is the final temperature of the mixture?

- A** 4.0°C
B 24.0°C
C 28.0°C
D 30.0°C
E 32.0°C

- 23** A 10 cm^3 sample of a gaseous hydrocarbon was completely burned in 80 cm^3 of oxygen gas (an excess) in a closed vessel.

The resulting mixture was cooled to the original room temperature and pressure, causing water vapour to condense completely. The remaining gas was then passed through an excess of aqueous potassium hydroxide, which absorbed all the carbon dioxide.

The volumes of gas measured at each stage, all under the same conditions of temperature and pressure, are shown in the table below:

Stage of experiment	Volume of gas / cm^3
Hydrocarbon vapour	10
Oxygen gas added initially (excess)	80
Gas mixture after combustion and cooling to room temperature	60
Gas mixture after treatment with excess aqueous potassium hydroxide	20

What is the molecular formula of the hydrocarbon?

- A** C_3H_6
B C_3H_8
C C_4H_6
D C_4H_8
E C_4H_{10}
- 24** An aqueous solution of an acid X forms a white precipitate when aqueous barium chloride, followed by dilute hydrochloric acid, is added to it.
- Acid X reacts with solid substance Y to form exactly two products. One of these products forms a green precipitate when aqueous sodium hydroxide is added to it.
- Which of the following could be X and Y?

	X	Y
A	HCl	CuO
B	HCl	Fe
C	HCl	FeCO_3
D	HCl	Zn
E	H_2SO_4	CuO
F	H_2SO_4	Fe
G	H_2SO_4	FeCO_3
H	H_2SO_4	Zn

- 25** Two monoprotic acids, X and Y , each have concentration $0.010 \text{ mol dm}^{-3}$ at the same temperature. The pH of acid X is 2 and the pH of acid Y is 3.

Which conclusion is best supported by this information?

- A** Acid X produces ten times the hydrogen-ion concentration of acid Y and is the stronger acid.
 - B** Acid Y produces ten times the hydrogen-ion concentration of acid X and is the stronger acid.
 - C** The acids have the same strength because their molar concentrations are equal.
 - D** Acid X produces twice the hydrogen-ion concentration of acid Y .
 - E** Acid Y does not dissociate in water.
- 26** A reaction between metallic zinc and dilute nitric acid produces an aqueous solution of zinc nitrate, water, and a gaseous oxide of nitrogen, X , as the only products.
- In the balanced chemical equation for this reaction, 4 mol of zinc reacts to produce 5 mol of water.
- What is the formula of the gaseous oxide of nitrogen X ?
- A** N_2O
 - B** NO
 - C** NO_2
 - D** N_2O_3
 - E** N_2O_5
- 27** Which of the following aqueous solutions has a pH of 1.0?
- A** 0.01 mol dm^{-3} hydrochloric acid
 - B** 0.1 mol dm^{-3} hydrochloric acid
 - C** 1.0 mol dm^{-3} hydrochloric acid
 - D** 0.1 mol dm^{-3} ethanoic acid
 - E** 0.1 mol dm^{-3} sodium hydroxide

Biology

- 1 A metabolic enzyme is located exclusively inside the cytoplasm of living cells. When this enzyme comes into direct contact with a colourless water-soluble substrate, it catalyses a reaction that produces a blue product.

Assume that neither the enzyme nor the substrate can cross an intact cell surface membrane.

Four separate suspensions were prepared by adding the substrate to:

- mammalian cells in an isotonic solution
- mammalian cells in pure water
- intact plant cells in pure water
- plant protoplasts (plant cells from which cell walls have been enzymatically removed) in pure water

All four mixtures were incubated at 25 °C for 10 minutes.

Which row in the table correctly predicts whether a blue colour will develop in each mixture?

	mammalian cells in isotonic solution	mammalian cells in pure water	intact plant cells in pure water	plant protoplasts in pure water
A	blue colour develops	no colour change	no colour change	blue colour develops
B	blue colour develops	blue colour develops	no colour change	no colour change
C	no colour change	blue colour develops	no colour change	blue colour develops
D	no colour change	blue colour develops	blue colour develops	no colour change
E	no colour change	no colour change	blue colour develops	blue colour develops
F	no colour change	blue colour develops	no colour change	no colour change
G	blue colour develops	no colour change	blue colour develops	no colour change
H	no colour change	no colour change	no colour change	blue colour develops

- 2 The diagram shows a simplified transverse section through a dicotyledonous plant leaf, highlighting two distinct cellular layers, P and Q.

Which of the following statements is/are correct?

- 1 Layer P and Layer Q are both tissues.
- 2 The leaf is classified as an organ system.
- 3 Mitochondria are present in the cytoplasm of cells in both Layer P and Layer Q.

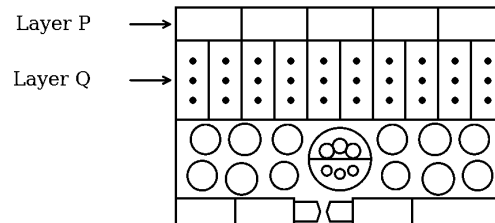


Diagram not to scale

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

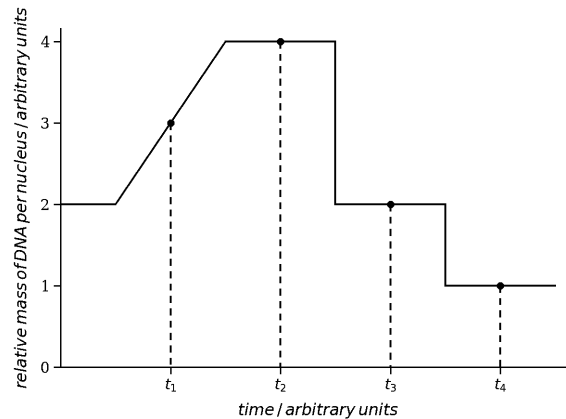
- 3** Which of the statements about the production of an insect-resistant transgenic crop plant are correct?
- 1 A restriction enzyme is used to cut the desired gene from the donor organism's DNA.
 - 2 DNA ligase is used to seal the gene into a plasmid vector.
 - 3 The genetically modified plant cells divide by meiosis during micropropagation to form new plantlets.
 - 4 All somatic cells in the mature regenerated plant will contain a copy of the inserted gene.
- A** 1 and 2 only
B 1 and 4 only
C 2 and 3 only
D 3 and 4 only
E 1, 2 and 4 only
F 1, 3 and 4 only
G 2, 3 and 4 only
H 1, 2, 3 and 4

- 4 The graph shows the relative mass of DNA per nucleus during meiosis and gamete formation in a diploid animal.

Four time points are labelled t_1 , t_2 , t_3 , and t_4 .

Which of the following statements is/are correct?

- 1 Semi-conservative replication of DNA occurs during the period containing time t_1 .
- 2 At time t_3 , the nucleus contains pairs of homologous chromosomes.
- 3 The number of chromosomes in a nucleus at time t_4 is equal to the number of chromosomes in a nucleus at time t_3 .



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 5** β -thalassaemia is an inherited condition in humans caused by a mutation in the gene coding for the β -globin polypeptide of haemoglobin, leading to a deficiency of functional haemoglobin.

In one form of gene therapy, haematopoietic stem cells are removed from the bone marrow of a patient with β -thalassaemia. A functional β -globin gene is inserted into the nuclear DNA of these stem cells. The genetically modified stem cells are then returned to the patient, where they establish in the bone marrow and can differentiate into red blood cells.

Which of the following statements is/are correct?

- 1 The modified stem cells can divide by mitosis to produce replacement stem cells that also contain the inserted gene.
- 2 Mature red blood cells circulating in the patient's blood will each contain a nucleus carrying the inserted gene.
- 3 In developing red blood cells, mRNA transcribed from the inserted gene is translated into functional β -globin polypeptides.

- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** 1 and 3 only
- F** 2 and 3 only
- G** 1, 2 and 3

- 6 The diagram represents the secondary structure of a transfer RNA (tRNA) molecule.

Which row in the table correctly identifies parts 1, 2, and 3, and describes the overall molecular nature of tRNA?

	Part 1	Type of bond at 2	Part 3	Overall structure
A	amino acid attachment site	phosphodiester	anticodon	single-stranded RNA
B	amino acid attachment site	hydrogen	codon	single-stranded RNA
C	amino acid attachment site	hydrogen	anticodon	double-stranded RNA
D	amino acid attachment site	hydrogen	anticodon	single-stranded RNA
E	ribosome attachment site	phosphodiester	codon	double-stranded RNA
F	ribosome attachment site	hydrogen	anticodon	single-stranded RNA
G	mRNA attachment site	hydrogen	codon	single-stranded RNA
H	mRNA attachment site	phosphodiester	anticodon	double-stranded RNA

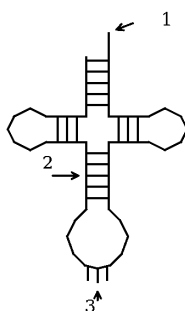


Diagram not to scale

- 7 The diagram shows the blood flow between four major mammalian organ systems, represented by boxes **W**, **X**, **Y**, and **Z**. Arrows indicate the direction of blood flow in the vessels connecting them.

Which row in the table correctly identifies boxes **W**, **X**, **Y**, and **Z**?

	Heart and lungs	Kidneys	Liver	Small intestine
A	X	Z	Y	W
B	Y	W	X	Z
C	Y	Z	W	X
D	Y	Z	X	W
E	Z	W	Y	X
F	Z	X	W	Y
G	X	W	Z	Y
H	Y	X	Z	W

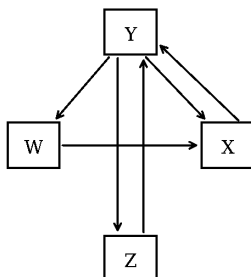


Diagram not to scale

- 8 An investigation was carried out using a single clone of the water flea *Daphnia magna* (all individuals were genetically identical).

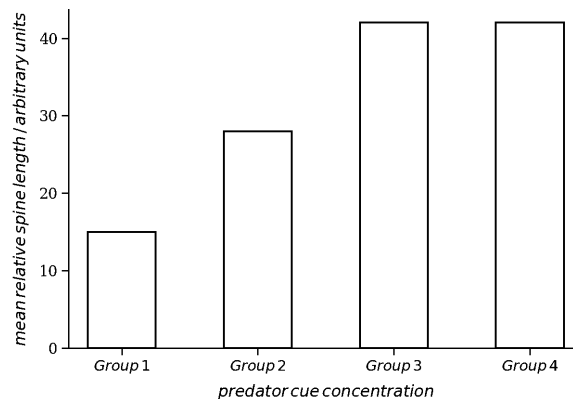
Forty newly hatched *Daphnia* of identical size were divided equally into four separate aquaria (Groups 1 to 4). The environmental conditions in the aquaria were controlled as follows:

- kept at the same constant water temperature
- supplied with the same daily quantity of algal food
- kept under the same light–dark cycle
- exposed to a different concentration of chemical cues (kairomones) released by predatory fish, where the concentration in Group 1 < Group 2 < Group 3 < Group 4

After reaching maturity, the relative length of the defensive tail spine was measured for each individual, and the mean relative spine length was calculated for each group. The results are shown in the graph below.

Which of the following statements could explain the results?

- 1 The difference in mean spine length between Group 1 and Group 2 is due to an environmental factor.
- 2 The mean spine length in Group 4 equals that in Group 3 because an environmental factor other than predator cue concentration is limiting growth.
- 3 The mean spine length in Group 4 equals that in Group 3 because the maximum spine length is constrained by the genotype of the clone.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 9 Red-green colour blindness is a sex-linked (X-linked) recessive condition in humans.

A survey of a group of 1000 students recorded the following data:

Group	Phenotype / status	Number of individuals
Males	Colour-blind	48
Males	Normal vision	552
Females	Colour-blind	4
Females	Carrier with normal vision	72
Females	Non-carrier with normal vision	324

One nucleated white blood cell was collected from every individual in the group.

What is the total number of red-green colour blindness alleles present across all of these collected cells?

- A 56
- B 124
- C 128
- D 176
- E 200
- F 248

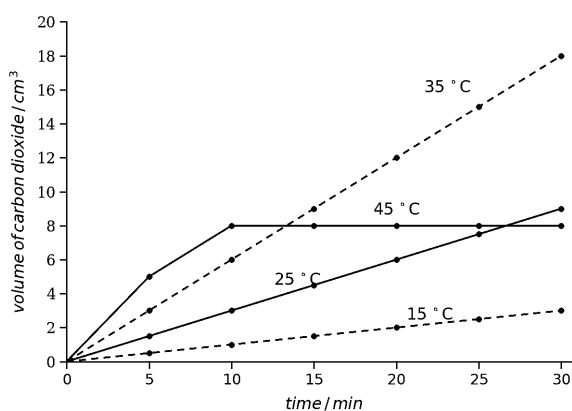
- 10** An investigation was carried out into the effect of temperature on anaerobic respiration in yeast.

A suspension of yeast cells was supplied with excess glucose solution and incubated at four different temperatures: 15 °C, 25 °C, 35 °C, and 45 °C. In each case, the cumulative volume of carbon dioxide released was recorded at regular intervals over a period of 30 minutes.

All other variables were kept constant. The results are shown on the graph below.

Which of the following statements is/are correct?

- 1 During the first 10 minutes, the average rate of carbon dioxide production was greatest at 45 °C.
- 2 For the reaction at 35 °C, the average rate of carbon dioxide production was $0.01 \text{ cm}^3 \text{ s}^{-1}$.
- 3 Between $t = 15$ minutes and $t = 30$ minutes, the rate of carbon dioxide production at 45 °C was greater than at 15 °C.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

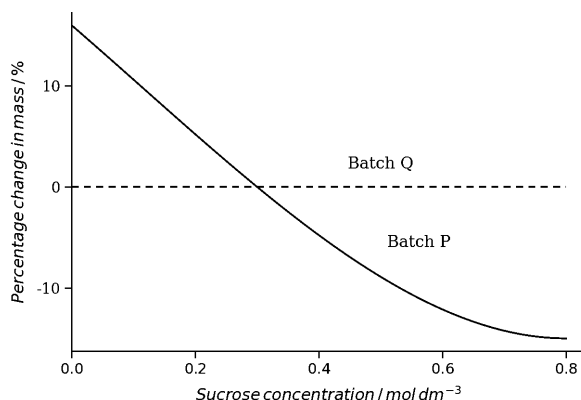
11 Identical cylinders of potato tissue were prepared and divided into two batches:

- Batch P consisted of untreated, living tissue.
- Batch Q was placed in boiling water for 10 minutes to kill the cells and then cooled to room temperature.

Both batches were immersed in sucrose solutions of concentrations ranging from 0.0 mol dm^{-3} to 0.8 mol dm^{-3} at 20°C for 2 hours. The percentage change in mass was measured, and the results are shown in the graph below.

Which of the following statements is/are correct?

- 1 At a sucrose concentration of 0.30 mol dm^{-3} , the water potential of the tissue in Batch P is equal to the water potential of the surrounding solution.
- 2 In Batch P, the pressure potential (Ψ_p) of the cells immersed in the 0.80 mol dm^{-3} sucrose solution is zero.
- 3 The lack of mass change in Batch Q is due to the breakdown of cellulose cell walls during boiling.



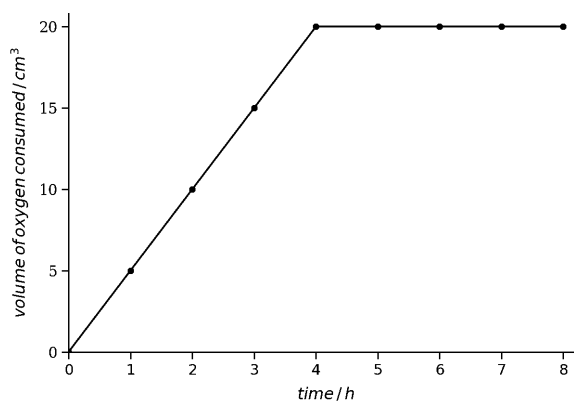
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 12 A student investigated the rate of aerobic respiration in germinating pea seeds using a sealed respirometer containing potassium hydroxide solution to absorb CO_2 . The respirometer contained 50 g of germinating seeds and was maintained at a constant temperature of 20°C .

The graph shows the cumulative volume of oxygen consumed over an 8-hour period.

Which row in the table gives the correct mean rate of oxygen consumption per gram of seeds between 0 and 4 hours, and the most likely reason why oxygen consumption stopped after 4 hours?

	Mean rate of oxygen consumption / $\frac{\text{cm}^3}{\text{g} \cdot \text{h}}$	Reason for oxygen consumption stopping after 4 h
A	0.050	available oxygen in the respirometer has been depleted
B	0.050	respiratory enzymes have denatured
C	0.10	available oxygen in the respirometer has been depleted
D	0.10	all respiratory enzyme active sites are saturated
E	0.40	available oxygen in the respirometer has been depleted
F	0.40	respiratory enzymes have denatured
G	5.0	available oxygen in the respirometer has been depleted
H	5.0	all respiratory enzyme active sites are saturated



- 13** Two stem cuttings taken from the same parent rose plant are grown in two different gardens. Which of the following could lead to phenotypic variation between the two mature plants?
- 1 differences in the intensity of light received by the plants
 - 2 crossing over during meiosis in the parent plant
 - 3 spontaneous mutations occurring during mitotic cell divisions
- A** none of them
 - B** 1 only
 - C** 2 only
 - D** 3 only
 - E** 1 and 2 only
 - F** 1 and 3 only
 - G** 2 and 3 only
 - H** 1, 2 and 3

- 14** An ecologist investigated the population size of an insect species in an isolated woodland using the mark-release-recapture method.

On Day 0, 200 insects were captured, marked on the thorax with waterproof paint, and released back into the woodland.

On each of the following five days (Days 1 to 5), a random sample of 80 insects was captured from the woodland. The number of marked individuals (R) in each sample was recorded, and all 80 insects were immediately released back into the habitat.

The graph shows the number of marked insects recaptured on each day.

Population size (N) can be estimated using the Lincoln index:

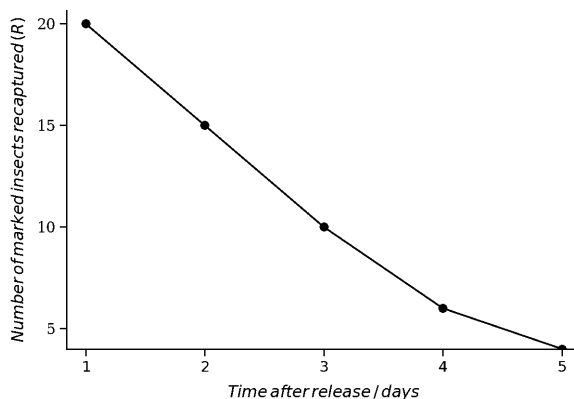
$$N = \frac{M \times C}{R}$$

where:

- M is the number of individuals initially marked on Day 0
- C is the total number of individuals captured in the recapture sample
- R is the number of marked individuals recaptured

Which of the following statements is/are correct?

- 1 Based on the sample collected on Day 1, the estimated population size in the woodland is 800.
- 2 If the paint made marked insects more conspicuous to predators, the Lincoln index would progressively overestimate the true population size over the five days.
- 3 If the insects became increasingly 'trap-shy' after their initial capture, the Lincoln index calculated on subsequent days would underestimate the true population size.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 15** The diagram shows a cross-section through part of a dicotyledonous leaf, showing different cellular layers.

Which of the following statements is/are correct?

- 1 The leaf is an organ, and Layer 1 and Layer 2 are each tissues.
- 2 The nucleus of a living cell in Layer 1 contains the genes that code for the enzymes required for chlorophyll synthesis.
- 3 When illuminated with white light, photophosphorylation occurs in the cells of both Layer 1 and Layer 2.

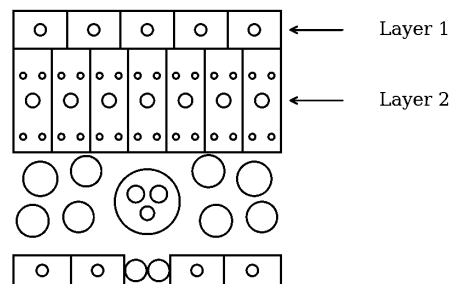


Diagram not to scale

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

16 The following statements describe various interspecific ecological interactions:

1 *Rhizobium* bacteria living inside root nodules of a legume, obtaining carbohydrates while supplying the plant with fixed nitrogen compounds

2 an epiphytic orchid growing on the high branch of a rainforest tree to access light, without absorbing nutrients or water from the tree

3 cleaner wrasse removing and feeding on ectoparasites from the skin and gills of a coral trout

4 mistletoe inserting specialized structures into the vascular tissues of a host tree to extract water and mineral ions

5 photosynthetic dinoflagellates (zooxanthellae) living inside reef-building coral cells, providing organic nutrients in exchange for inorganic compounds and shelter

Which of the statements describe(s) an example of mutualism?

- A** 1 only
- B** 3 only
- C** 1 and 5 only
- D** 2 and 4 only
- E** 3 and 5 only
- F** 1, 2 and 3 only
- G** 1, 3 and 5 only
- H** 2, 4 and 5 only

- 17 The enzyme β -galactosidase catalyses the hydrolysis of ortho-nitrophenyl- β -D-galactopyranoside (ONPG) into galactose and ortho-nitrophenol (ONP).

* ONPG is colourless in solution.

* ONP is colourless at neutral and acidic pH, but turns bright yellow under alkaline conditions (absorbance measured at 420 nm).

Five test tubes were prepared containing equal volumes and concentrations of ONPG and β -galactosidase at pH 7.0. Each tube was incubated at a different temperature for 10 minutes.

After 10 minutes, an equal volume of concentrated Na_2CO_3 solution was added to each tube to raise the pH to 10.5. The absorbance of each solution at 420 nm was measured immediately and then measured again after 10 minutes at room temperature.

Tube	Incubation temperature / $^{\circ}\text{C}$	Absorbance at 420 nm immediately after Na_2CO_3 addition	Absorbance at 420 nm after 10 minutes
1	10	0.14	0.14
2	25	0.41	0.41
3	40	0.86	0.86
4	55	0.35	0.35
5	70	0.03	0.03

Which of the following statements is/are correct?

- 1 The addition of Na_2CO_3 solution halted the catalytic activity of the β -galactosidase.
- 2 At the end of the initial 10-minute incubation, the concentration of unreacted substrate remaining was lowest in tube 3.
- 3 The optimum temperature for this β -galactosidase must be 40°C .

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

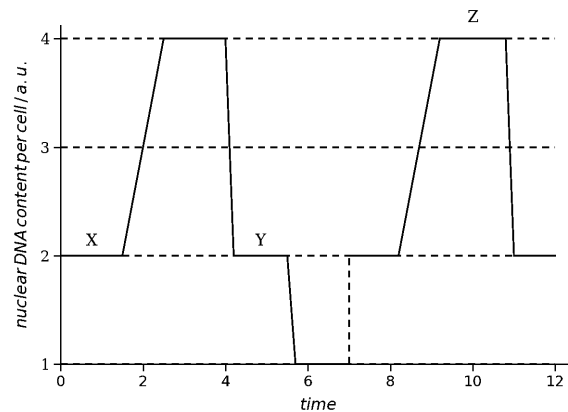
- 18** The graph shows changes in the nuclear DNA content of a single cell lineage during spermatogenesis in a human male, followed by fertilisation and the first mitotic division of the resulting zygote.

The male is heterozygous (Aa) for an autosomal gene. During meiosis I, the chromosome carrying allele A segregates into the cell lineage shown. The resulting mature sperm cell fertilises an ovum from a female who is homozygous (AA) for this gene.

Assume that no crossing over or mutations occur.

How many copies of allele A are present in the single cell at each of the stages marked **X**, **Y**, and **Z** on the graph?

	stage X	stage Y	stage Z
A	1	1	2
B	1	1	4
C	1	2	2
D	1	2	4
E	2	1	2
F	2	1	4
G	2	2	2
H	2	2	4



- 19 A culture of yeast cells is suspended in an aqueous nutrient medium.

The table below shows the concentrations of two dissolved solutes, substance **P** and substance **Q**, as well as the total solute concentration, in the nutrient medium and in the cytoplasm of the yeast cells.

	Concentration in medium / mmol dm^{-3}	Concentration in cytoplasm / mmol dm^{-3}
substance P	0.2	5.0
substance Q	15.0	1.5
total solutes	30.0	180.0

Under normal aerobic conditions, there is a net movement of substance **P**, substance **Q**, and water molecules into the yeast cells.

The culture is then treated with a metabolic poison that completely prevents the synthesis of ATP.

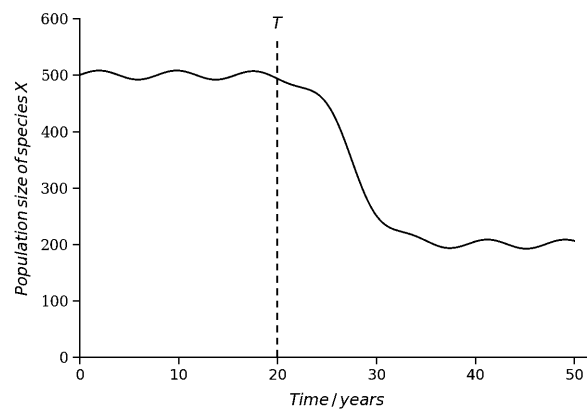
Which row in the table below correctly shows the net movement of substance **P**, substance **Q**, and water molecules across the cell membrane immediately after the addition of this poison?

	Net movement of substance P	Net movement of substance Q	Net movement of water
A	moves into the cell	moves into the cell	moves into the cell
B	moves into the cell	moves into the cell	moves out of the cell
C	moves into the cell	no net movement into the cell	moves into the cell
D	moves into the cell	no net movement into the cell	moves out of the cell
E	no net movement into the cell	moves into the cell	moves into the cell
F	no net movement into the cell	moves into the cell	moves out of the cell
G	no net movement into the cell	no net movement into the cell	moves into the cell
H	no net movement into the cell	no net movement into the cell	moves out of the cell

- 20** The graph shows changes in the population size of a small rodent species (species X) on an isolated island over a 50-year period. At time T , a non-native mammal species (species Y) was introduced to the island.

Which of the following could account for the change in the population of species X observed after time T ?

- 1 Species Y competed with species X for some, but not all, of its food resources.
- 2 Species Y was competitively superior and occupied the exact same ecological niche as species X.
- 3 Species Y preyed on species X, but species X had access to a limited number of safe refuges.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 21** An intact, living plant cell is surrounded by an aqueous solution.

Which of the following will **always** result in a net movement of water into the cell by osmosis?

1 transferring the cell into a solution that has a higher water potential than the solute potential of the cell's cytoplasm

2 increasing the concentration of dissolved solutes inside the cell's central vacuole

3 increasing the water potential of the surrounding solution to a value greater than the total water potential of the cell

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

- 22** During hot, dry conditions, a terrestrial plant closes its stomata while remaining in bright daylight.

Which of the following could be a direct or indirect consequence of this stomatal closure?

1 A decrease in the rate of transpiration from the leaves.

2 A reduction in the concentration gradient of carbon dioxide between the surrounding air and the leaf air spaces.

3 An increase in the temperature of the leaf tissue.

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

- 23** An intracellular enzyme, enzyme X, is present in the cytoplasm of both mammalian cells and *Escherichia coli* (*E. coli*) bacteria. Enzyme X catalyses the conversion of a colourless substrate, S, into a fluorescent product, P.

Neither enzyme X nor substrate S can cross an intact cell surface membrane.

Lysozyme is an enzyme that digests the peptidoglycan cell wall of bacteria.

Four separate suspensions were prepared:

1. Mammalian cells suspended in pure water
2. Intact *E. coli* bacteria suspended in pure water
3. *E. coli* bacteria pre-treated with lysozyme to remove their cell walls, then suspended in pure water
4. *E. coli* bacteria pre-treated with lysozyme to remove their cell walls, then suspended in an isotonic buffer solution

Substrate S was added to each suspension.

Which row in the table correctly shows whether fluorescent product P will be detected in each mixture?

Key:

✓ = product P detected

X = product P not detected

	Mammalian cells in pure water	Intact <i>E. coli</i> in pure water	Lysozyme-treated <i>E. coli</i> in pure water	Lysozyme-treated <i>E. coli</i> in isotonic buffer
A	✓	X	✓	X
B	✓	✓	✓	X
C	✓	X	X	X
D	✓	X	✓	✓
E	X	X	✓	X
F	X	✓	X	✓
G	✓	✓	X	X
H	X	X	X	X

- 24** The epithelial lining of the mammalian small intestine undergoes continuous renewal. Adult stem cells located at the base of intestinal crypts divide, and their daughter cells differentiate as they migrate along the villi.

A healthy adult has an estimated total intestinal epithelial surface area of 0.30 m^2 .

A biopsy shows that there are an average of 4.0×10^5 epithelial cells per mm^2 of this surface.

The total number of epithelial cells is maintained at a constant level, and the average lifespan of an epithelial cell is 5.0 days (120 hours).

Which of the following statements is/are correct?

- 1 The average rate of new epithelial cell production is 1.0×10^9 cells per hour.
- 2 The stem cells at the base of the crypts divide by meiosis to maintain the cell population.
- 3 Mitochondria in mature intestinal epithelial cells produce the ATP required for the active transport of nutrients.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 25** Three stages of aerobic cellular respiration in human cells, X, Y and Z, are compared in the table below:

stage of respiration	releases CO ₂	directly consumes molecular O ₂	produces reduced NAD (NADH)	synthesises ATP
X	no	no	yes	yes
Y	yes	no	yes	yes
Z	no	yes	no	yes

Which row in the table below correctly identifies stages X, Y and Z?

	X	Y	Z
A	glycolysis	Krebs cycle	oxidative phosphorylation
B	glycolysis	oxidative phosphorylation	Krebs cycle
C	Krebs cycle	glycolysis	oxidative phosphorylation
D	Krebs cycle	oxidative phosphorylation	glycolysis
E	oxidative phosphorylation	glycolysis	Krebs cycle
F	oxidative phosphorylation	Krebs cycle	glycolysis

- 26** Consider the following three cell types:

- a bacterium (*Escherichia coli*)
- a unicellular fungus (yeast, *Saccharomyces cerevisiae*)
- a plant palisade mesophyll cell

Which of the following features is/are present in **all three** of these cell types?

- 1 circular DNA
- 2 70S ribosomes
- 3 membrane-bound organelles
- 4 enzyme-catalysed synthesis of ATP

- A** 4 only
- B** 1 and 2 only
- C** 1 and 4 only
- D** 2 and 4 only
- E** 3 and 4 only
- F** 1, 2 and 4 only
- G** 2, 3 and 4 only
- H** 1, 2, 3 and 4

27 Which of the following mature cell types contain a nucleus?

- 1 mature companion cells in phloem
- 2 mature human red blood cells
- 3 mature phloem sieve tube elements
- 4 mature leaf guard cells

- A** 1 and 2 only
- B** 1 and 4 only
- C** 2 and 3 only
- D** 2 and 4 only
- E** 3 and 4 only
- F** 1, 2 and 3 only
- G** 1, 3 and 4 only
- H** 1, 2, 3 and 4